

On the Turán Number of the Blow-Up of the Hexagon

Taiki Aiba

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Abstract

The r -blowup of a graph F , denoted by $F[r]$, is the graph obtained by replacing the vertices and edges of F with independent sets of size r and copies of $K_{r,r}$, respectively. In this article, we follow the survey by Janzer, Methuku, and Nagy [2] and prove that $\text{ex}(n, C_6[2]) = O(n^{5/3})$ and for any positive integer t , $\text{ex}(n, \theta_{3,t}[2]) = O(n^{5/3})$, which is tight when t is sufficiently large.

1 Introduction

The extremal number of a graph H , denoted $\text{ex}(n, H)$, is the maximum possible number of edges in a graph not containing H as a not necessarily induced subgraph on n vertices. We wish to construct bounds on the extremal number for graphs known as *theta graphs* and the *2-blow-up* of them.

Given a graph G , we define the r -blow-up of a graph, denoted $G[r]$, to be the graph formed by replacing each vertex of G with an independent set of size r and each edge of G with a copy of $K_{r,r}$ between the vertices of the independent sets represented by the endpoints of the edge in G .

Grzesik, Janzer, and Nagy made the following conjecture [1]:

Conjecture 1. *Let F be a graph such that $\text{ex}(n, F) = O(n^{2-\alpha})$ for some constant $0 \leq \alpha \leq 1$. Then for any positive integer r ,*

$$\text{ex}(n, F[r]) = O(n^{2-\frac{\alpha}{r}}).$$

These authors also proved this conjecture in the case that F is a tree and in the case when $F = K_{s,t}$ and $\alpha = \frac{1}{s}$, which covers $C_4 = K_{2,2}$. This led to the curious question, stated as a conjecture as follows [1]:

Conjecture 2. *For any $r, k \geq 2$,*

$$\text{ex}(n, C_{2k}[r]) = O(n^{2-\frac{1}{r}+\frac{1}{rk}}).$$

We define the theta graph, denoted $\theta_{k,t}$, as a graph with endpoints u, v and t pairwise internally vertex-disjoint (u, v) -paths each of length k (here, length is defined by the number of edges in the path).

In this paper, we will prove that Conjecture 2 holds for $r = 2$ and $k = 3$. In fact, we will prove the following more general result [2]:

Theorem 1. *For any positive integer t ,*

$$\text{ex}(n, \theta_{3,t}[2]) = O(n^{5/3}).$$

Note that Conjecture 2 subject to $r = 2$ and $k = 3$ is a specific case of Theorem 1 with $t = 2$.

2 Proof of Theorem 1

We will first state an equivalent result to Theorem 1. We define a K -almost-regular graph G as a graph that satisfies $\max_{v \in V(G)} d_G(v) \leq K \cdot \min_{v \in V(G)} d_G(v)$. Jiang and Seiver proved the following [3]:

Theorem 2. *Let ε, c be positive reals, where $\varepsilon < 1$ and $c \geq 1$. Let n be a positive integer that is sufficiently large as a function of ε . Let G be a graph on n vertices with $e(G) \geq cn^{1+\varepsilon}$. Then G contains a K -almost-regular subgraph G' on $m \geq n^{\frac{\varepsilon-\varepsilon^2}{2+2\varepsilon}}$ vertices such that $e(G') \geq \frac{2c}{5}m^{1+\varepsilon}$ and $K = 20 \cdot 2^{\frac{1}{\varepsilon^2}+1}$.*

It is also well-known that any graph G contains a bipartite subgraph with at least $e(G)/2$ edges. Thus, Theorem 1 reduces to the following:

Theorem 3. *Let K be a constant, and let G be a K -almost-regular bipartite graph on n vertices with minimum degree $\delta = \omega(n^{2/3})$. Then, for n sufficiently large, G contains a copy of $\theta_{3,t}[2]$.*

The proof of Theorem 3 is split into four parts: bounding the number of quadruples with a large common neighborhood, bounding the number of triples with a large common neighborhood, finding $P_3[2]$'s with prescribed extra properties, and counting the number of pairs of $P_3[2]$'s that share an internal vertex.

Throughout this proof, we shall refer to the following figure:

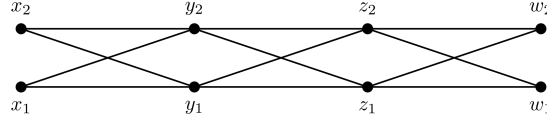


Figure 1 [2]. A copy of $P_3[2]$.

Henceforth, any mention of the vertices x_i, y_i, z_i, w_i for $i \in [2]$ refers to the labeling of these vertices in Figure 1.

2.1 Bounding the number of quadruples with a large common neighborhood

Given a graph, we wish to classify copies of $P_3[2]$ where both $d(\{x_1, x_2, z_1, z_2\})$ and $d(\{y_1, y_2, w_1, w_2\})$ are sufficiently small. The reason that we wish to consider these quadruples is because even if we have plenty of copies of $P_3[2]$ in a graph and we try to take the union of t of them, it may still be degenerate; i.e., some of the internal vertices may coincide in the t copies of $P_3[2]$.

We will accomplish the goal of limiting the number of quadruples with a large common neighborhood by proving the following lemma:

Lemma 1. *Let G be a $\theta_{3,t}[2]$ -free graph. Let x, x', y, y' be distinct vertices in G , and let $R \subset N(\{y, y'\}) \setminus \{x, x'\}$. Then the number of ordered pairs of distinct vertices $(z, z') \in R^2$ with $d(\{x, x', z, z'\}) \geq 6t$ is at most $4t|R|$.*

Proof. Take a set $S = \{(z_1, z'_1), \dots, (z_s, z'_s)\} \in R^2$ that is maximal with respect to size such that $z_1, z'_1, \dots, z_s, z'_s$ are pairwise distinct and $d(\{x, x', z_i, z'_i\}) \geq 6t$ for all $i \in [s]$. If $s \geq t$, then we may choose $w_1, w'_1, \dots, w_t, w'_t \in V(G)$ such that $x, x', y, y', z_i, z'_i, w_j, w'_j$ for all $i, j \in [t]$ are pairwise distinct, and $w_i, w'_i \in N(\{x, x', z_i, z'_i\})$ for all $i \in [t]$. But $x, x', y, y', z_i, z'_i, w_j, w'_j$ for all $i, j \in [t]$ form a $\theta_{3,t}[2]$, a contradiction.

Thus, $s < t$. By the maximality of S , for all $(z, z') \in R^2$ such that $d(\{x, x', z, z'\}) \geq 6t$, we have $\{z, z'\} \cap \{z_1, z'_1, \dots, z_s, z'_s\} \neq \emptyset$. This leaves at most $2 \cdot 2s \cdot |R| < 4t|R|$ possibilities for such (z, z') because there are 2 ways to order the pair, $2s$ ways to have the first vertex in the pair be in $\{z_1, z'_1, \dots, z_s, z'_s\}$ and $|R|$ ways to have the second vertex be in R . $\square$

2.2 Bounding the number of triples with a large common neighborhood

This step is a key idea in this proof. Indeed, for a fixed pair (z'_1, z'_2) of vertices with $d(\{z'_1, z'_2\}) \approx n^{1/3}$, if there are too many (e.g., greater than $n^{1/3}$) vertices z_1 with $d(\{z_1, z'_1, z'_2\}) \gg n^{1/6}$, then we can find a bipartite subgraph B with sufficiently many edges, and a result by Grzesik, Janzer, and Nagy [1] implies that we can find the 2-blow-up of any tree in B . In particular, we can find the 2-blow-up of a path of length 4 whose degree 2 vertices are embedded to $N(\{z'_1, z'_2\})$, forming $C_6[2]$ as a subgraph. For general t , we simply need to embed a larger 2-blow-up of B to obtain a $\theta_{3,t}[2]$ subgraph.

We will first state without proof the following preliminary lemma:

Lemma 2. *Let T be a tree, and let $v \in V(T)$. Then there exists a constant $C = C(T)$ with the following property. Let H be a bipartite graph with parts X and Y such that $d(y) \geq C|X|^{1/2}$ for every $y \in Y$. Assume that the number of triples of distinct vertices $(y, x_1, x_2) \in Y \times X \times X$ with $yx_1, yx_2 \in E(H)$ is more than $C|X|^2$. Then H contains a copy of $T[2]$ with the two images of v embedded in X .*

The proof may be found in the original paper and uses a bound on the number of $K_{2,2}$ subgraphs and probabilistic methods [2]. We will use this lemma to prove the following lemma:

Lemma 3. *There exists a constant $C = C(t)$ with the following property. Let G be a $\theta_{3,t}[2]$ -free bipartite graph. Let z_1, z_2 be distinct vertices in G , and let $N(\{z_1, z_2\})$ have size ℓ . Let $R = \{v \in V(G) \setminus \{z_1, z_2\} : d(\{v, z_1, z_2\}) \geq C\ell^{1/2}\}$. Then the number of triples (z', w_1, w_2) of distinct vertices with $z' \in R$, $w_1, w_2 \in N(\{z', z_1, z_2\})$ is at most $C\ell^2$.*

Proof. Let C be sufficiently large and suppose that the number of triples (z', w_1, w_2) of distinct vertices with $z' \in R$, $w_1, w_2 \in N(\{z', z_1, z_2\})$ is more than $C\ell^2$. By Lemma 2 with $X = N(\{z_1, z_2\})$, $Y = R$, and $H = G[X \cup Y]$, there exist pairwise distinct vertices $u, u', w_1, w'_1, \dots, w_t, w'_t \in N(\{z_1, z_2\})$ and $v_1, v'_1, \dots, v_t, v'_t \in R$ such that $v_i, v'_i \in N(\{u, u', w_i, w'_i\})$ for all $i \in [t]$. Then the vertices $z_1, z_2, u, u', w_1, w'_1, \dots, w_t, w'_t, v_1, v'_1, \dots, v_t, v'_t$ form a copy of $\theta_{3,t}[2]$, a contradiction.

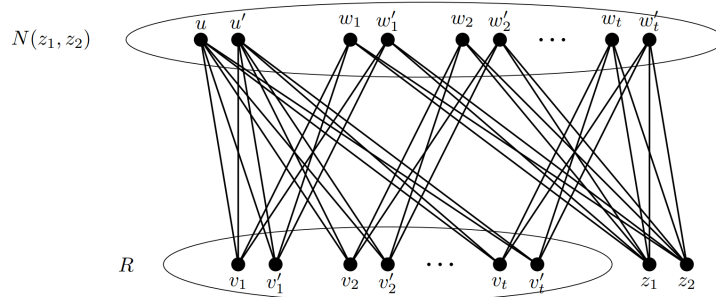


Figure 2 [2]. A copy of $P_3[2]$ found within this configuration.

□

2.3 Finding $P_3[2]$'s with prescribed extra properties

Now, we will state without proof a lemma that finds the number of tuples (y_1, y_2, z_1, z_2) given a fixed ordered pair (x_1, x_2) that characterizes copies of $P_3[2]$ that we wish to have:

Lemma 4. *Let K be a constant, and let G be a K -almost-regular, $\theta_{3,t}[2]$ -free graph on n vertices with minimum degree $\delta = \omega(n^{2/3})$. Then there exist distinct vertices x_1, x_2 in G and a set $S \subset N(\{x_1, x_2\})$ of size at least $n^{1/3}$ as follows. Writing $s = |S|$, there exist $\lambda = \omega(1)$, $\mu = \omega(1)$, and a collection $\mathcal{Q}$ of $\Omega(\frac{s^2 n^{2/3} \lambda^{27/10}}{\mu^{11/10}})$ tuples $(y_1, y_2, z_1, z_2) \in V(G)^4$ satisfying the following properties:*

1. $y_1, y_2 \in S$ and $y_i z_j$ are edges for every i, j .
2. $x_1, x_2, y_1, y_2, z_1, z_2$ are pairwise distinct.
3. $d_S(z_1), d_S(z_2) \leq \lambda \frac{s}{n^{1/3}}$.
4. $d(\{x_1, x_2, z_1, z_2\}) < 6t$.
5. $\mu n^{1/3} \leq d(\{z_1, z_2\}) \leq 2\mu n^{1/3}$.

The proof may be found in the original paper and uses degrees of vertices, intricate bounding arguments, and involved computations [2].

From here, we will only need to include w_1, w_2 , which we will do in the next subsection.

2.4 Counting the number of pairs of $P_3[2]$'s that share an internal vertex

In this final part of the proof, we will extend our result from Lemma 4 to include w_1, w_2 . We will start with the following lemma, which uses Lemma 1 and property 5 of Lemma 4 in its proof [2]:

Lemma 5. *Let G be a $\theta_{3,t}[2]$ -free bipartite graph on n vertices. Let x_1, x_2 be distinct vertices, let $S \subset N(\{x_1, x_2\})$ be a set of size $s \geq n^{1/3}$, and let $\lambda, \mu = \omega(1)$. Assume that there is a set $\mathcal{Q}$ of $q = \Omega(\frac{s^2 n^{2/3} \lambda^{27/10}}{\mu^{11/10}})$ tuples $(y_1, y_2, z_1, z_2) \in V(G)^4$ satisfying the five properties in Lemma 4.*

Then there exists a set $\mathcal{A}$ of $\Omega(\mu^4 n^{-2/3} q^2)$ tuples $(y_1, y_2, z_1, z_2, w_1, w_2, y'_2, z'_1, z'_2) \in V(G)^9$ for which the following properties hold:

1. $(y_1, y_2, z_1, z_2), (y_1, y'_2, z'_1, z'_2) \in \mathcal{Q}$.
2. $w_1, w_2 \in N(\{z_1, z_2, z'_1, z'_2\})$.
3. $d(\{y_1, y_2, w_1, w_2\}), d(\{y_1, y'_2, w_1, w_2\}) < 6t$.
4. $x_1, x_2, y_1, y_2, z_1, z_2, w_1, w_2, y'_2, z'_1, z'_2$ are pairwise distinct.

We will now state two more lemmas without proof that, when combined with Lemmas 4 and 5, will prove Theorem 3 and consequently, will prove Theorem 1 [2]:

Lemma 6. *In the setting of Lemma 5, the number of tuples*

$$(y_1, y_2, z_1, z_2, w_1, w_2, y'_2, z'_1, z'_2) \in \mathcal{A}$$

with $d(\{z_1, z'_1, z'_2\}) \geq \mu^{7/5} n^{1/6}$ is $o(|\mathcal{A}|)$. Moreover, the number of tuples $(y_1, y_2, z_1, z_2, w_1, w_2, y'_2, z'_1, z'_2) \in \mathcal{A}$ with $d(\{z_2, z'_1, z'_2\}) \geq \mu^{7/5} n^{1/6}$ is also $o(|\mathcal{A}|)$.

The proof of this lemma makes use of Lemma 3 and property 5 of Lemma 4 since the bound on the number of triples (z_1, z'_1, z'_2) and (z_2, z'_1, z'_2) with a large common neighborhood will also bound the number of tuples $(y_1, y_2, z_1, z_2, w_1, w_2, y'_2, z'_1, z'_2)$ with this undesired property.

The second lemma is the following:

Lemma 7. *In the setting of Lemma 5, there are $o(|\mathcal{A}|)$ tuples $(y_1, y_2, z_1, z_2, w_1, w_2, y'_2, z'_1, z'_2) \in \mathcal{A}$ for which $d(\{z_1, z'_1, z'_2\}) < \mu^{7/5} n^{1/6}$ and $d(\{z_2, z'_1, z'_2\}) < \mu^{7/5} n^{1/6}$.*

Finally, we may combine Lemmas 4, 5, 6, and 7 to prove Theorem 3. The authors Janzer, Methuku, and Nagy do not detail this combination of results, and it is left as an exercise to the reader.

References

- [1] A. Grzesik, O. Janzer, and Z. Nagy. *The Turán number of blow-ups of trees*. 2019. arXiv: 1904.07219 [math.CO]. URL: <https://arxiv.org/abs/1904.07219>.
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